

Lecture 11

Chapter 28



Kirchhoff's Laws



Gustav Robert Kirchhoff
(1824-1887)



Course website:

http://faculty.uml.edu/Andriy_Danylov/Teaching/PhysicsII

While still only a graduate student, he published a paper that included a pair of rules for the analysis of circuits (Kirchhoff's laws of circuits).

Today we are going to discuss:

Chapter 28:



- *Section 28.2 Kirchhoff's Laws*
- *Example 28.10 Analyzing a two-loop circuit*



Kirchhoff's Rules



Gustav Robert Kirchhoff
(1824-1887)

Idea! Some circuits are too complicated to analyze
(none of the elements are in series/parallel)

Kirchhoff's rules are very helpful.

To analyze a circuit means to find:

1. ΔV across each component
2. The current in each component



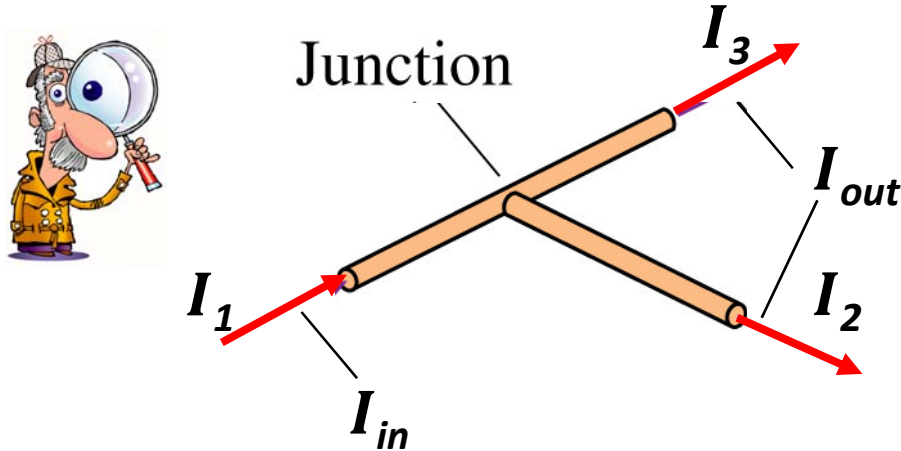
Kirchhoff's Junction Law

For a junction, the *law of conservation of current* requires that:

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$



At any junction point, the sum of all currents entering the junction must equal the sum of all currents leaving the junction.



$$\text{Junction law: } I_1 = I_2 + I_3$$



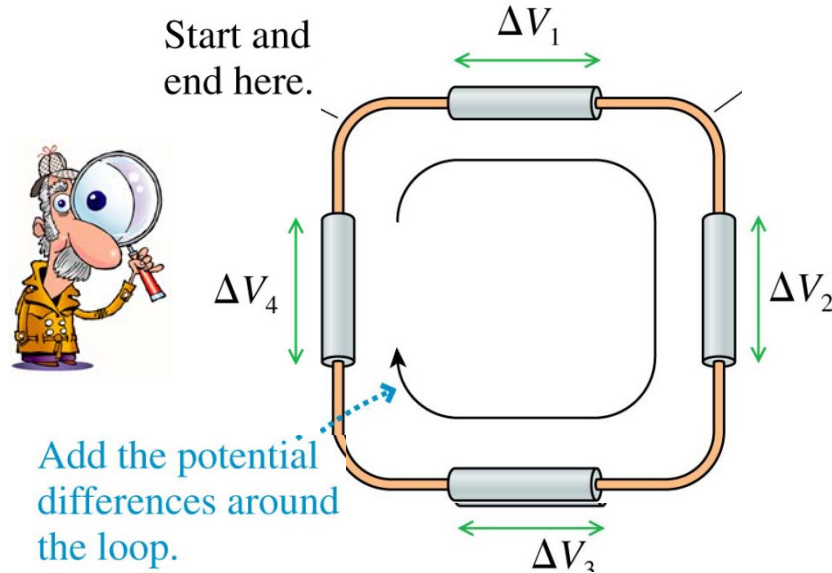
Kirchhoff's Loop Law

For any path that starts and ends at the same point:

$$\Delta V_{\text{loop}} = \sum (\Delta V)_i = 0$$



The sum of all the potential differences encountered while moving around a loop or closed path is zero.



$$\text{Loop law: } \Delta V_1 + \Delta V_2 + \Delta V_3 + \Delta V_4 = 0$$

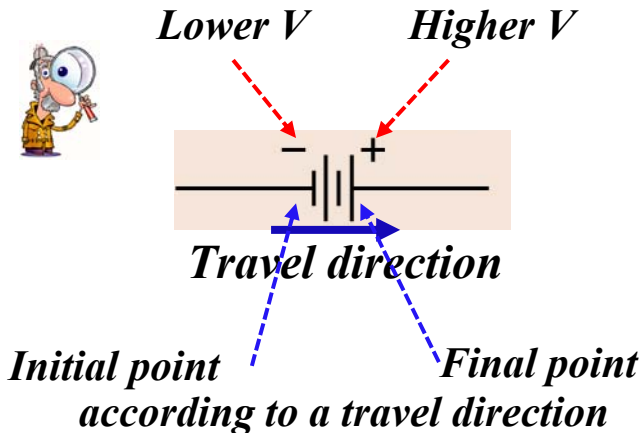
Now, we need to learn how to calculate these ΔV . Let's start with a battery:



ΔV across a battery



according to a property of a battery

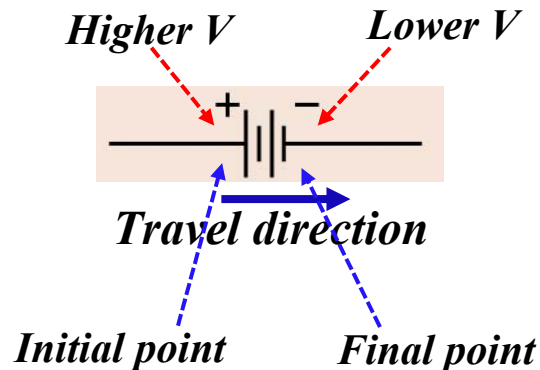


$$\Delta V = V_f - V_i = V_+ - V_- \Rightarrow \Delta V > 0 \quad \text{Potential increases}$$

■ For an ideal battery in the negative-to-positive direction:

$$\Delta V_{\text{bat}} = +\mathcal{E}$$

For a battery, the potential difference is positive if your chosen loop direction is from the negative terminal toward the positive terminal



$$\Delta V = V_f - V_i = V_- - V_+ \Rightarrow \Delta V < 0 \quad \text{Potential decreases}$$

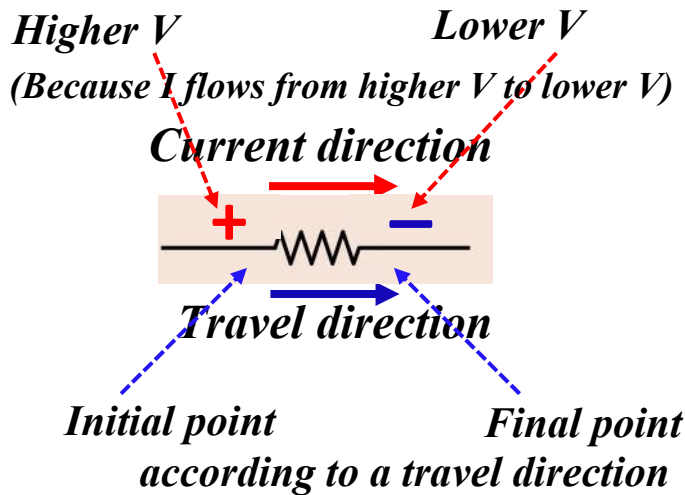
■ For an ideal battery in the positive-to-negative direction:

$$\Delta V_{\text{bat}} = -\mathcal{E}$$

The potential difference is negative if the loop direction is from the positive terminal toward the negative terminal



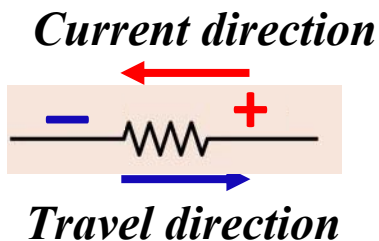
ΔV 's across resistors



Potential decreases

$$\Delta V = V_f - V_i = V_- - V_+ = -IR$$

For a resistor, apply Ohm's law; the potential difference is negative (a decrease) if your **chosen loop direction is the same as the chosen current direction** through that resistor



Potential increases

$$\Delta V = V_f - V_i = V_+ - V_- = +IR$$

For a resistor, apply Ohm's law; the potential difference is positive (an increase) if your **chosen loop direction is opposite to the chosen current direction** through that resistor

Example **Example 28.1.**
Analyze the circuit

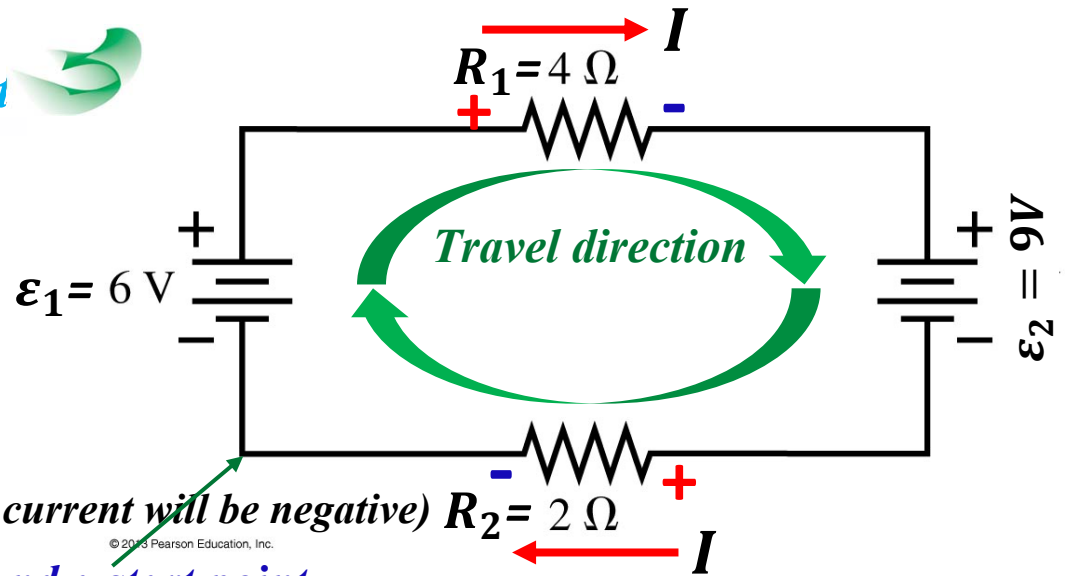
$$\sum I_{\text{in}} = \sum I_{\text{out}} \quad \text{No junction points}$$

$$\sum (\Delta V)_i = 0 \quad \text{Loop rule}$$

1) Assume CW direction of current

(If our assumption turns out to be wrong, the current will be negative)

2) Choose a travel direction (say, CW) and a start point



$$\sum (\Delta V)_i = 0 \quad \Rightarrow \quad +\varepsilon_1 - IR_1 - \varepsilon_2 - IR_2 = 0$$

$$I = \frac{\varepsilon_1 - \varepsilon_2}{R_1 + R_2}$$

Now we can find pot. differences across each resistor

$$\Delta V_{R_1} = IR_1 = \left(\frac{\varepsilon_1 - \varepsilon_2}{R_1 + R_2} \right) R_1$$

$$\Delta V_{R_2} = IR_2 = \left(\frac{\varepsilon_1 - \varepsilon_2}{R_1 + R_2} \right) R_2$$

ConcepTest

What is ΔV across the unspecified circuit element?

Loop rule



A) 0V

B) 1V

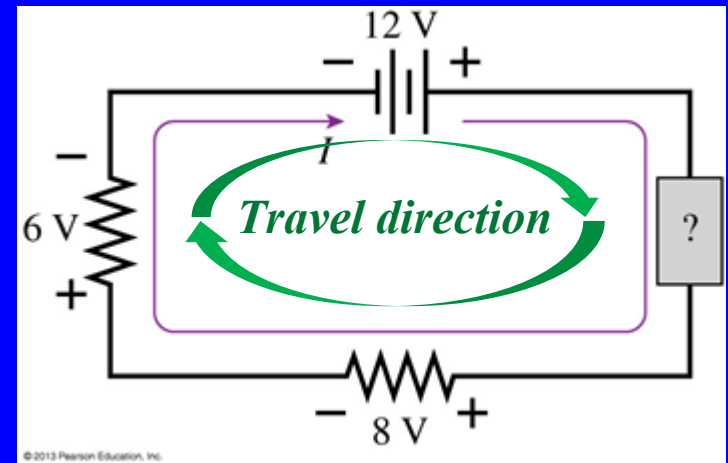
C) 2V

D) 3V

$$\Delta V_{\text{loop}} = \sum (\Delta V)_i = 0$$

$$+12\text{ V} + \Delta V - 8\text{ V} - 6\text{ V} = 0$$

$$\Delta V = 2\text{ V}$$

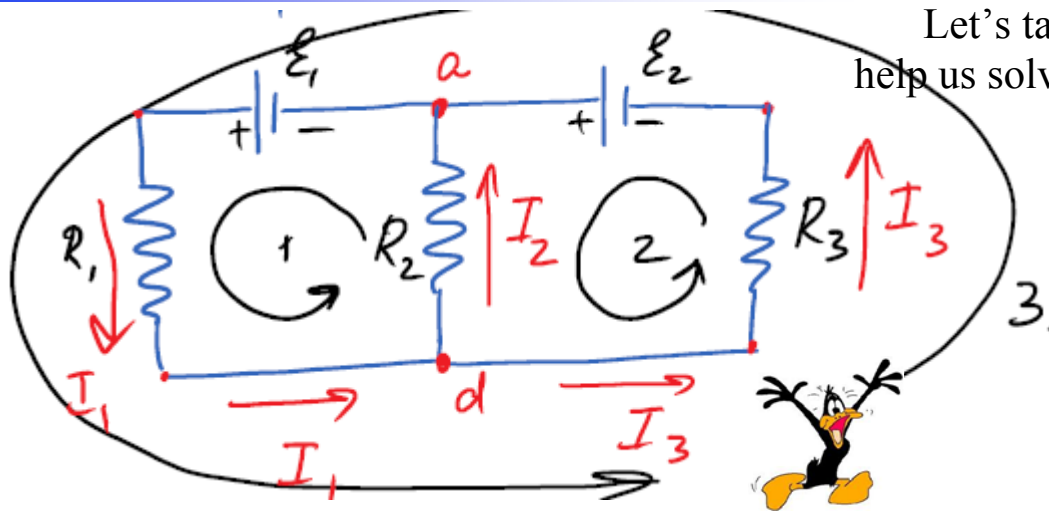


© 2013 Pearson Education, Inc.

Example Multi-Loop Circuit



Let's take a look at how the junction rule and loop rule help us solve for the unknown values in multi-loop circuits.



In general: if there are N junctions in a circuit, then there are $N-1$ independent junction equations

There are three legs between two junction points a, d
 1) we need to pick I_1, I_2, I_3 for each leg.
 (Direction is arbitrary)

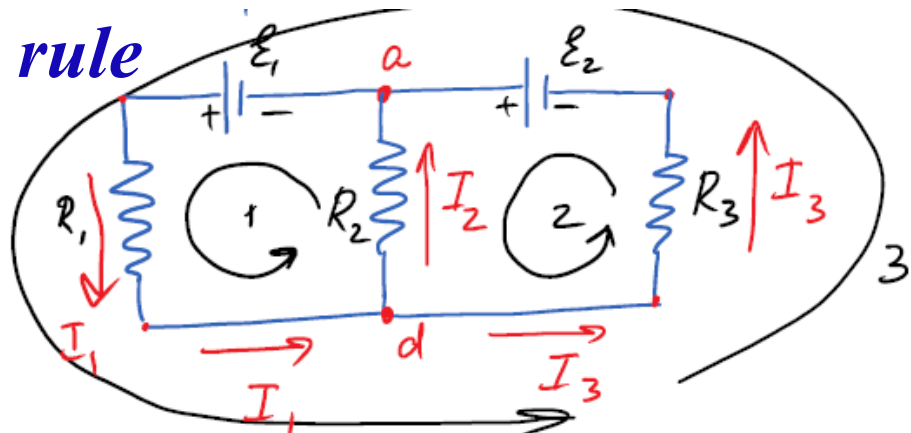
• Junction rule

for a : $I_3 + I_2 = I_1$

for d : $I_1 = I_2 + I_3$ (1)

You see, they are not independent (the same)
 so, we take one of them.

Loop rule



$$\begin{array}{c} \text{Travel direction} \\ \Delta V_{\text{res}} = -IR \end{array}$$

$$\begin{array}{c} \text{Travel direction} \\ \Delta V_{\text{res}} = IR \end{array}$$

There are three loops in the circuit, so we can write 3 loop equations.

$$\text{loop 1: } +\mathcal{E}_1 - I_1 R_1 - I_2 R_2 = 0 \quad \checkmark$$

$$\text{loop 2: } +\mathcal{E}_2 + I_2 R_2 - I_3 R_3 = 0 \quad \checkmark$$

$$\text{loop 3: } +\mathcal{E}_2 + \mathcal{E}_1 - I_1 R_1 - I_3 R_3 = 0 \quad \times$$

So, finally

$$\begin{cases} I_1 = I_2 + I_3 \\ \mathcal{E}_1 - I_1 R_1 - I_2 R_2 = 0 \\ \mathcal{E}_2 + I_2 R_2 - I_3 R_3 = 0 \end{cases}$$

Now see $\mathcal{E}q 3 = \mathcal{E}q 1 + \mathcal{E}q 2$, so only two loop eq-ns are independent. Let's take $\mathcal{E}q 1$ and 2

3 eq-ns + 3 unknowns
Physics is over. The rest is algebra..

Tactics: Using Kirchhoff's Rules

1. Label the current in each separate branch of the given circuit with a different subscript, such as I_1, I_2, I_3 . Each current refers to a segment between two junctions. Choose the direction of each current, using an arrow. The direction can be chosen arbitrarily: if the current is actually in the opposite direction, it will come out with a minus sign in the solution.

2. Identify the unknowns. You will need as many independent equations as there are unknowns. You may write down more equations than this, but you will find that some of the equations will be redundant (that is, not be independent in the sense of providing new information). You may use $\mathcal{E}$ for each resistor, which sometimes will reduce the number of unknown

3. Apply Kirchhoff's junction rule at one or more junctions.

3. Apply Kirchhoff's loop rule for one or more loops: follow each loop in one direction only. Pay careful attention to subscripts, and to signs:

(a) For a resistor, apply Ohm's law; the potential difference is negative (a decrease) if your chosen loop direction is the same as the chosen current direction through that resistor; the potential difference is positive (an increase) if your chosen loop direction is opposite to the chosen current direction.

(b) For a battery, the potential difference is positive if your chosen loop direction is from the negative terminal toward the positive terminal; the potential difference is negative if the loop direction is from the positive terminal toward the negative terminal.

4. Solve the equations algebraically for the unknowns.

Thank you
See you next time

